

8(a)	any seven from: 1 photosynthetic / named / primary / accessory, pigments to absorb light (energy) ; 2 ref. to, photosystems / antenna complex and reaction centre / light harvesting structures ; 3 photoactivation / electrons excited / emission of electrons ; 4 electrons move along electron transport chain ; 5 (cyclic / non-cyclic) photophosphorylation / light-dependent stage ; 6 site of photolysis / location of oxygen-evolving complex ; 7 thylakoids stacked to form grana ; 8 gives large surface area ; 9 thylakoid space or lumen to, form proton gradient / have high concentration of protons ; 10 thylakoid membrane is (relatively) impermeable, to maintain the proton gradient ; 11 ATP synthase to make ATP; 12 chemiosmosis ;	7
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Question	Answer	Marks
8(b)	any four from: <i>describe</i> 1 A activity increases and B activity remains, low / constant ; 2 comparative data quote ; e.g. 3.0–3.2 at 0.5 mins and 11.4–11.6 at 6 mins for A 0.4 throughout for B <i>max three:</i> <i>explanation:</i> 3 RA / rubisco activase, <u>activates</u> rubisco ; 4 by changing the active site of rubisco ; 5 by enabling rubisco to bind more readily with, RuBP / substrate or more enzyme–substrate complexes ; 6 enables products to leave active site more quickly ; 7 AVP ; e.g. cofactor action qualified	4

Question	Answer	Marks
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